Backtracking Algorithms

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Backtracking Algorithm

A backtracking algorithm works by recursively exploring all possible solutions to a problem. It starts by choosing an initial solution, and then it explores all possible extensions of that solution. If an extension leads to a solution, the algorithm returns that solution.

If an extension does not lead to a solution, the algorithm backtracks to the previous solution and tries a different extension.

Often applied in constraint satisfaction problems like N-Queens Problem,Graph Coloring, puzzles and path finding.

Backtracking Algorithm

Backtracking Algorithm Steps :

Step -1: Choose an initial solution.

Step -2: Explore all possible extensions of the current solution.

Step -3: If an extension leads to a solution, return that solution.

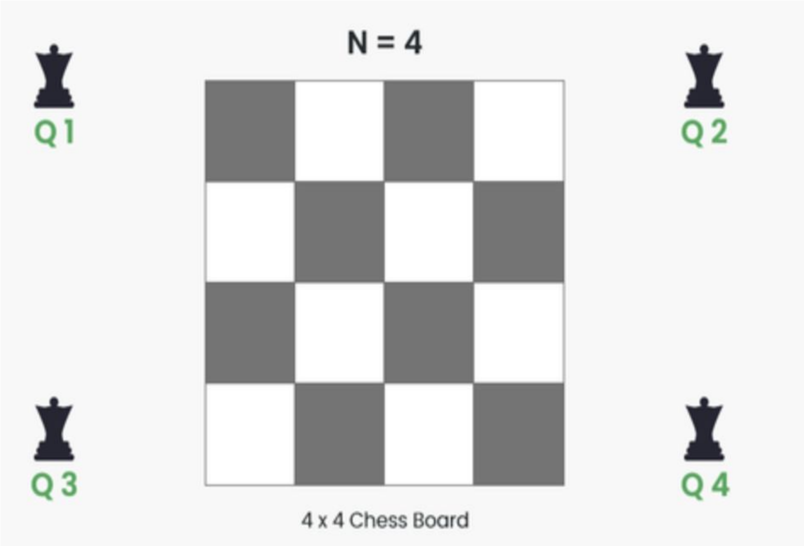
Step -4: If an extension does not lead to a solution, backtrack to the previous solution and try a different extension.

Step -5: Repeat steps 2-4 until all possible solutions have been explored.

Example of Backtracking Algorithm

N Queen Problem

Given an integer n, the task is to find the solution to the n-queens problem, where n queens are placed on an n*n chessboard such that no two queens can attack each other.



N Queen Problem Solution

Rules for Placing Queens

Each queen can attack another queen if they are in the same:

1.Row

2.Column

3.Diagonal

The challenge is to place N queens in such a way that no queens attack each other.

N Queen Problem Solution

Step-1: Start from the first row and try placing a queen in each column.

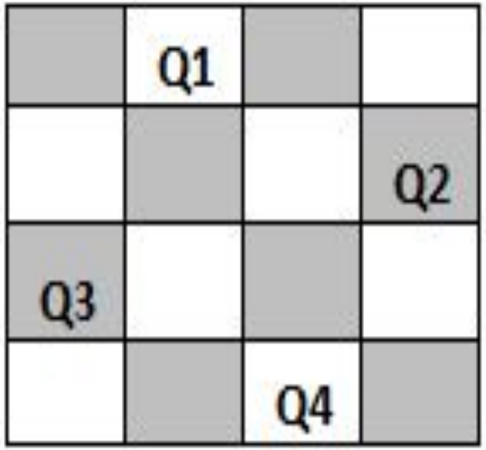
Step-2: Check if the placement is safe (i.e., no other queens threaten it).

Step-3: If safe, place the queen and move to the next row.

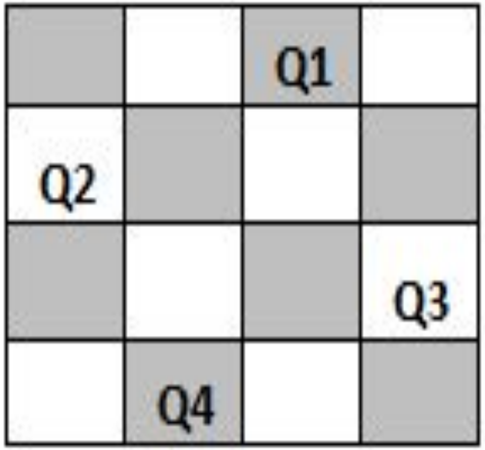
Step-4: If a row has no safe column, backtrack (remove the previous queen and try a different column).

Step-5: Repeat until all queens are placed or all options are exhausted.

N Queen Problem Solution



Solution 1



Solution 2

Complexity of Backtracking Algorithm

Highly dependent on the problem.

Worst case: $O(b^d)$ where b is branching factor and d is depth of recursion.

Advantage of Backtracking Algorithm

1.Backtracking can be used to solve complex combinatorial optimization problems. It can be adapted to various constraints and problem-specific requirements.

2.Backtracking can be optimized through pruning.

3.Backtracking algorithms can be parallelized and executed in multiple threads.

Disadvantage of Backtracking Algorithm

1.Backtracking can be very slow for large problems.

2.Backtracking can also be memory-intensive, as it requires storing the state of the partial solution at each step of the search.

3.Backtracking can be inefficient, as it may explore the same parts of the search space multiple times.